Closed-form 2nd-order Data Shapley as a Spurious-Correlation Attenuator

Where it works, where it doesn't — and why

A closed-form, single-pass per-sample reweighter for worst-group robustness.

Mechanism reproduced across **three** spurious regimes: color, texture, natural images.

The problem: spurious correlations break worst-group accuracy

- Models latch onto an **easy spurious cue** (background, color, texture) that correlates with the label in training but **flips at test time**.
- Average accuracy looks fine; the **worst group collapses**.
- Strong fixes need **group labels** (GroupDRO) or expensive two-stage/bi-level pipelines (JTT, meta-reweighting).

> **Question:** can a **closed-form, single-pass** per-sample reweighter — no group labels at training, no inner optimization loop — attenuate the spurious cue?

Our answer

Repurpose **In-Run Data Shapley** (Wang et al. 2024) as an in-training reweighter, and add a **2nd-order gradient–Hessian–gradient cross-term**. We isolate exactly what the cross-term contributes and show it generalizes across three spurious modalities.

Idea: Data Shapley as an in-training reweighter

- In-Run Data Shapley gives a **closed-form Taylor estimate** of how much each training sample improves a small balanced validation anchor — computed **during** training, no retraining.
- We turn that valuation into a **per-sample weight** and run weighted SGD.
- A 50:50 group-balanced **anchor** is used only for the inner inner-product — sensitive labels are never consumed by the model update.

$$w_i = B \cdot \mathrm{softmax}(\ \mathrm{EMA}(\varphi_i) / \tau\ ) \quad \rightarrow \quad \theta_{t+1} = \theta_t - \eta \textstyle\sum_i w_i \nabla L(z_i)$$

The whole question becomes: **what should φ_i be?**

Method — 1st order (Fairds-1)

$$\varphi_i^{(1)} = \langle\, g_i, g_{val} \,\rangle \quad g_i = \nabla_\theta L(z_i),\ \ g_{val} = \nabla_\theta L(D_{val})$$

- Wang et al.'s estimator: how much sample z_i reduces validation loss in one SGD step.
- Larger $\varphi^{(1)} \Rightarrow z_i$ pushes θ in a validation-helpful direction.
- Per-sample gradients via **torch.func.vmap** in a single autograd graph — closed form, no bi-level loop.

> But the 1st-order term alone leaves accuracy on the table. Can curvature help?

Method — 2nd order (Fairds-2) with RMS rescaling

$$\varphi_i^{(2)} = \varphi_i^{(1)} - \alpha \cdot \langle\, g_i,\, H_{val}\, g_{val} \,\rangle \quad \text{(cross-term)}$$

- $H_{val}\, g_{val}$ via the **Pearlmutter trick** — one extra backward pass, no full Hessian.
- H_{val}'s spectrum varies by orders of magnitude during training ⇒ a fixed α is unusable.
- **Algorithmic contribution:** rescale the cross-term to the **same RMS magnitude** as the 1st-order term each minibatch.

⚠ **Without RMS rescaling, Fairds-2 collapses to chance under strong spurious bias.**

Is the 2nd-order term doing something real?

A skeptic's worry: maybe the cross-term is just **smoothing / regularization** noise. We tested this directly.

What we found about the geometry

- g_{val} is almost the **top eigenvector of H_{val}** (alignment 0.89–0.99) ⇒ the cross-term is **nearly parallel** to $\varphi^{(1)}$.
- Decompose: $\text{cross}_n = \boldsymbol{\beta\cdot\varphi^{(1)}}$ (parallel, a curvature shrinkage) + $\mathbf{r}$ (orthogonal residual).

> The parallel part = global shrinkage (equivalent to smoothing).
> **Is the orthogonal residual r a real, sample-specific signal — or noise?**

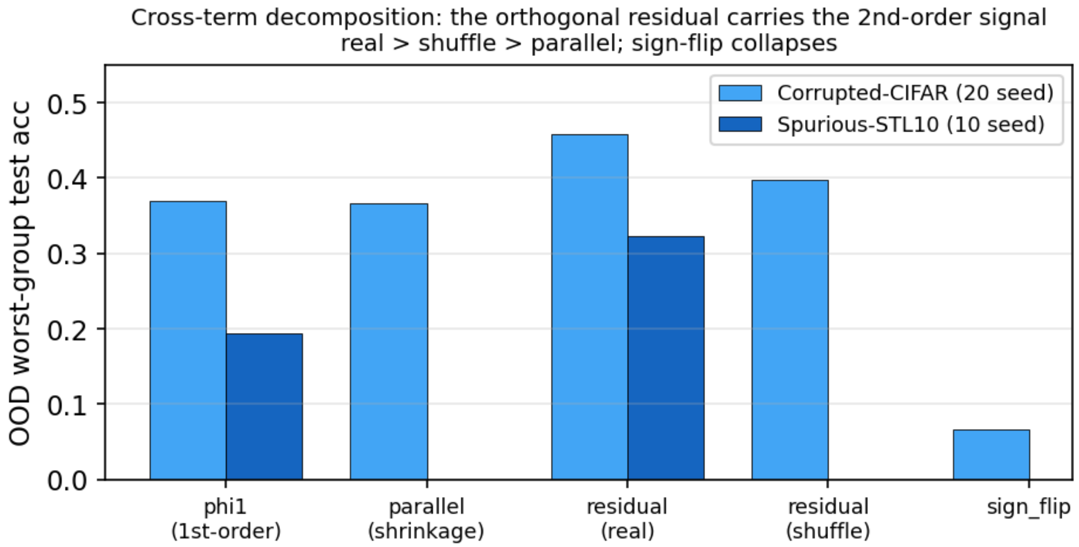
Residual ablation: the cross-term's signal lives in *r*

We build 5 arms that share $\varphi^{(1)}$ but treat *r* differently:

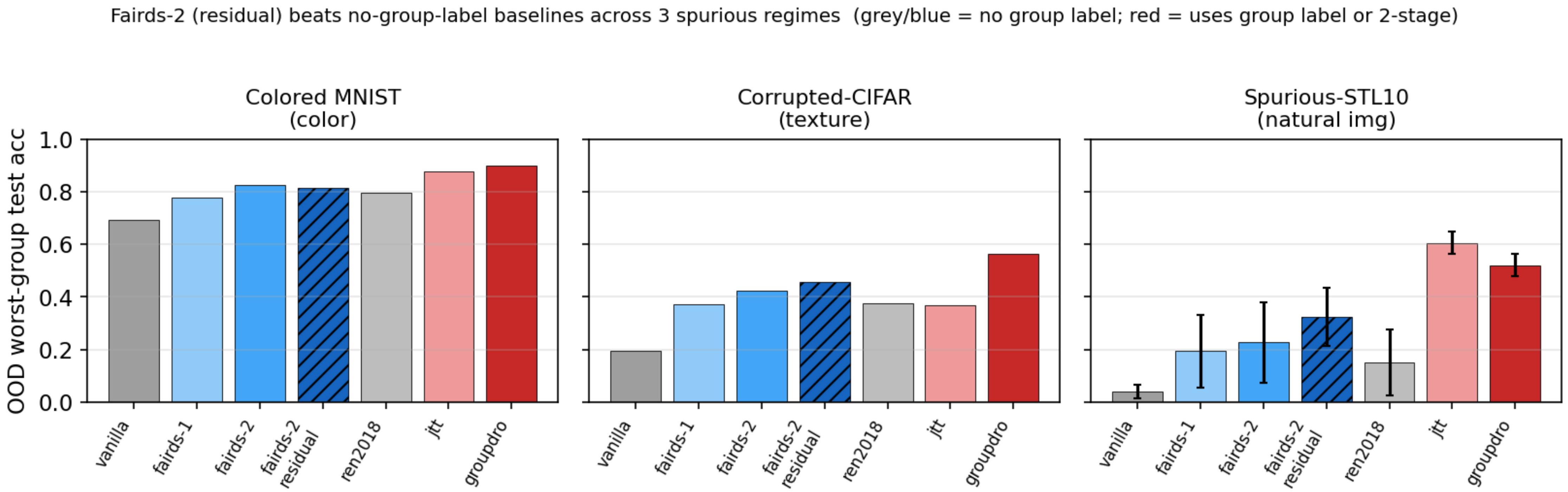
- **residual (real):** $\varphi^{(1)} - \alpha \cdot r$
- **residual (shuffle):** r permuted within class×φ-bin
- **parallel:** only the shrinkage part
- **sign_flip:** $\varphi^{(1)} + \alpha \cdot r$

real > shuffle > parallel;
sign-flip collapses.

CIFAR: real vs shuffle **+5.9pp, p=0.0015** (20 seeds). If *r* were noise, shuffling it would not matter — it does.



Headline: the mechanism reproduces across 3 spurious regimes



Fairds-2 (residual) beats every **no-group-label** baseline (vanilla, Fairds-1, Ren2018) in **all three** regimes — color (MNIST), texture (CIFAR), and natural images (STL-10).

The third regime (STL-10, 96px natural images)

New benchmark: car vs truck, corruption-type (blur vs noise) as the spurious cue; from-scratch CNN, 10 seeds.

method	OOD test_worst	vs Fairds-2-resid
vanilla	0.042	**+28.1pp, p=0.002**
Fairds-1 (1st-order)	0.194	**+12.9pp, p=0.010**
Ren2018 (bi-level)	0.151	**+17.2pp, p=0.002**
Fairds-2 (residual)	**0.323**	—
JTT / GroupDRO*	0.61 / 0.52	**stronger (see next)**

2nd-order isolation holds: the residual variant beats the 1st-order term by +12.9pp (p=0.010) — exactly the cross-term effect we isolated on CIFAR, now in a third modality.

Honest scope — what we do *not* claim

Stronger methods exist (reported plainly)

- **JTT** (2-stage ERM) and **GroupDRO** (uses group labels) reach higher worst-group accuracy in most regimes.
- Fairds-2's value is the **closed-form, single-pass cross-term mechanism**, not peak accuracy.

A regime where it fails: pretrained-ResNet Waterbirds fine-tuning

- All Fairds variants (3 algorithmic versions) **fail to match** the bi-level baseline.
- Cause: with a strong pretrained extractor, early per-sample gradients are small/noisy ⇒ the EMA buffer fills with noise. **Use bi-level meta-learning there.**

> We characterize the **regime boundary** rather than hide the negative result.

Cost & practicality

Wall-clock overhead (ResNet-18)

method	×vanilla
Fairds-1	4.30×
Ren2018 (bi-level)	4.78×
Fairds-2	7.68×

1st-order is cheaper than bi-level; 2nd-order pays for the HVP. Honestly reported, not yet at the 1.5×/3× target.

When to reach for Fairds-2

- **From-scratch / small models** with a strong spurious cue and no group labels.
- When you want a **closed-form** reweighter (no inner loop) and lowest seed variance among no-label methods.
- **Not** for pretrained fine-tuning ⇒ use bi-level.

Contributions

- **1.** Repurpose closed-form In-Run Data Shapley as an in-training per-sample reweighter for worst-group robustness.
- **2.** A **2nd-order cross-term** with **RMS rescaling** (required for stability under strong spurious bias).
- **3.** A **decomposition + residual ablation** proving the cross-term carries a real, sample-specific directional signal (real > shuffle > parallel; sign-flip collapses).
- **4.** The mechanism **reproduces across three spurious regimes** — color, texture, natural images.
- **5.** An **honest regime boundary**: a clean negative on pretrained fine-tuning, with a diagnosis.

Takeaway

> A **closed-form 2nd-order Data Shapley** reweighter attenuates spurious correlations across **three distinct regimes** — and we can prove *which part of the cross-term* does the work.

- The gain is the orthogonal **residual** of the gradient–Hessian–gradient term, not mere smoothing.
- It beats every no-group-label baseline from scratch; group-label / 2-stage methods remain stronger.
- It fails on pretrained fine-tuning — and we say so, with a mechanism.

Closed-form valuation can shape training dynamics — within a characterized regime.